

GO\_Biological\_Process\_2023 Top pathways by non-permutation

| Geneset                                   | stat        | num.genes | pval      | p.adj     | gene.vals                                                                                                |
|-------------------------------------------|-------------|-----------|-----------|-----------|----------------------------------------------------------------------------------------------------------|
| 'De Novo' AMP Biosynthetic Process (GO:0  | -0.0825843  | 4         | 5.674e-01 | 9.886e-01 | PFAS.111 PAICS.7003.5 ATIC.7003.5 ADSS1.7003.5 NA NA NA                                                  |
| 'De Novo' Post-Translational Protein Fol  | 0.13105097  | 15        | 7.895e-02 | 9.886e-01 | DNAJB13:142 SDF2L1:644 HSPA8:799 HSPA14:923 PTGES3:5662.5 HSPA2:5662.5 DNAJB14:5662.5                    |
| 2-Oxoglutarate Metabolic Process (GO:000  | 0.08438378  | 13        | 2.922e-01 | 9.886e-01 | GOT1:14 OGDHL:660 GOT2:675 IDH1:5662.5 PHYH:5662.5 PTHYH:5662.5 GPT2:5662.5                              |
| 3'-UTR-mediated mRNA Destabilization (GO  | -0.02439411 | 11        | 7.816e-01 | 9.886e-01 | DND1:1094 ZFP36:2084 TRIM71:2003.5 RC3H1:7003.5 ZFP93L1:7003.5 ZFP36L2:7003.5 MOV10:7003.5               |
| 3'-UTR-mediated mRNA Stabilization (GO:0  | -0.06303912 | 11        | 4.692e-01 | 9.886e-01 | ELAVL4:1331 YBX3:1592 MYD88:204 FZL9:7003.5 TARDPB:7003.5 MAPK14:7003.5 DAZL:7003.5                      |
| 3'-Phosphoadenosine 5'-Phosphosulfate Me  | 0.10660456  | 8         | 2.965e-01 | 9.886e-01 | SULT1A2:323 PAPSS2:5662.5 TPT32:5662.5 SULT2B1:5662.5 SULT1B1:5662.5 SULT1E1:5662.5 ENPPI1:5662.5        |
| 5S Class rRNA Transcription By RNA Polym  | -0.23755463 | 5         | 6.584e-02 | 9.886e-01 | GTF3C5:373 GTF3C5:378 GTF3C4:1652 GTF3C2:7003.5 GTF3C6:7003.5 NA NA                                      |
| 7-Methylguanosine RNA Capping (GO:000945  | 0.26546942  | 2         | 1.935e-01 | 9.886e-01 | RNMT:311 CMTR2:5662.5 NA NA NA NA NA                                                                     |
| 7-Methylguanosine mRNA Capping (GO:00063  | 0.26546942  | 2         | 1.935e-01 | 9.886e-01 | RNMT:311 CMTR2:5662.5 NA NA NA NA NA                                                                     |
| ADP Transport (GO:0015866)                | -0.12074398 | 5         | 3.498e-01 | 9.886e-01 | SLC25A24:813 SLC25A23:2170 SLC25A42:7003.5 SLC25A41:7003.5 SLC25A5:7003.5 NA NA                          |
| AMP Biosynthetic Process (GO:0006167)     | -0.11099817 | 6         | 3.464e-01 | 9.886e-01 | PFAS:111 APRT:1396 PAICS:7003.5 ATIC:7003.5 ADSS1:7003.5 NUDT2:7003.5 NA                                 |
| AMP Metabolic Process (GO:0046033)        | 0.04567587  | 9         | 6.352e-01 | 9.886e-01 | AK9:324 ADSS1:5662.5 NUDT2:5662.5 NTF5C1A:5662.5 NTF5E:5662.5 AMPD1:5662.5 AMPD3:5662.5                  |
| ATF6-mediated Unfolded Protein Response   | 0.05376563  | 5         | 6.772e-01 | 9.886e-01 | MBTPS1:5662.5 XBP1:5662.5 ATF6B:5662.5 DDIT3:5662.5 CREBZF:5662.5 NA NA                                  |
| ATP Biosynthetic Process (GO:0006754)     | 0.07835106  | 13        | 3.281e-01 | 9.886e-01 | ATP5F1C:339 ATP5PD:423 NUDT2:5662.5 TGFBI:5662.5 VPS9D1:5662.5 ATP5PF:5662.5 ATP5PB:5662.5               |
| ATP Metabolic Process (GO:0046034)        | -0.02145636 | 25        | 7.105e-01 | 9.886e-01 | PKRKA2:478 NADK:762 PARP1:1028 FIGLN1:1715 MYH8:1858 CTNS:2217 ENPPI1:7003.5                             |
| ATP Synthase Coupled Electron Transport   | 0.04792873  | 3         | 7.737e-01 | 9.886e-01 | NUDFB6:717 NUDFS:5662.5 NUDFV3:10845 NA NA NA NA                                                         |
| ATP Transport (GO:0015867)                | -0.09707394 | 9         | 3.133e-01 | 9.886e-01 | SLC25A24:813 CALHM1:893 SLC25A23:2170 SLC25A42:7003.5 SLC25A41:7003.5 SLC25A5:7003.5 CALHM3:7003.5       |
| AV Node Cell Action Potential (GO:000601  | -0.17229959 | 6         | 1.439e-01 | 9.886e-01 | CACNB2:335 SCN4B:1328 CACNA1G:2205 SCN10A:2193 ARPC5:7003.5 SCN5A:7003.5 TRPM4:7003.5 NA                 |
| AV Node Cell To Bundle Of His Cell Commu  | 0.05376016  | 4         | 7.096e-01 | 9.886e-01 | SCN5A:5662.5 CXADR:5662.5 GJC3:5662.5 GJA5:5662.5 NA NA NA NA                                            |
| Arp2/3 Complex-Mediated Actin Nucleation  | -0.04453935 | 10        | 6.258e-01 | 9.886e-01 | WASHC1:820 ARPC1A:977 ARPC5L:2053 WHAMM:2179 ARPC5:7003.5 ARPC3:7003.5 JMY:7003.5                        |
| B Cell Activation (GO:0042113)            | 0.09162123  | 61        | 1.374e-02 | 9.886e-01 | HDAC9:43 CD86:62 FLT3:159 CASP8:207 CD40:419 CD79A:527 MEFC2:761                                         |
| B Cell Activation Involved In Immune Res  | 0.04962331  | 12        | 4.761e-01 | 9.886e-01 | RNF168:118 MSH2:5662.5 GPR183:5662.5 PCDH9:5662.5 CD180:5662.5 LPGA:5662.5 RNF8:5662.5                   |
| B Cell Differentiation (GO:0030183)       | 0.08932514  | 44        | 4.058e-02 | 9.886e-01 | HDAC9:43 FLT3:159 CD79A:527 PTPRC:780 IL4:881 SLC25A5:5662.5 UTPN2:5662.5                                |
| B Cell Proliferation (GO:0042100)         | 0.0628080   | 15        | 3.166e-02 | 9.886e-01 | CD40:419 CD79A:527 MEFC2:761 PTPRC:780 IL4:881 SLC25A5:5662.5 CD70:5662.5 IFNE:5662.5                    |
| B Cell Receptor Signaling Pathway (GO:00  | 0.09914482  | 26        | 8.032e-02 | 9.886e-01 | CD79A:527 CTLA4:566 MEFC2:761 PTPRC:780 NFAM1:5662.5 LAMA5:295 JAG2:319 SYBU:742                         |
| BMP Signaling Pathway (GO:0030509)        | -0.01336705 | 42        | 7.646e-01 | 9.886e-01 | FAM83C:157 INHBE:62 LEFTY1:835 DDX5:960 MSAS4:983 ACVR2A:1596 LEFTY2:1837                                |
| C-Terminal Protein Amino Acid Modificati  | -0.01019528 | 8         | 3.180e-01 | 9.886e-01 | GPLD1:832 AGBL5:2195 AGBL4:2243 ATGPBP1:7003.5 FOLH1:7003.5 FOLH1:7003.5 ATG7:7003.5                     |
| C-Terminal Protein Deglutamylation (GO:0  | -0.13526423 | 4         | 3.488e-01 | 9.886e-01 | AGBL5:2195 AGBL4:2243 ATGPBP1:7003.5 FOLH1:7003.5 NA NA NA NA                                            |
| C-Terminal Protein Lipidation (GO:000650  | -0.10932515 | 3         | 3.120e-01 | 9.886e-01 | GPLD1:832 ATG7:7003.5 ATG7:7003.5 NA NA NA NA NA                                                         |
| C4-dicarboxylate Transport (GO:0015740)   | 0.03951597  | 10        | 6.653e-01 | 9.886e-01 | SLC25A12:782 SLC25A13:5662.5 SLC13A2:5662.5 SLC25A18:5662.5 SLC13A5:5662.5 SLC1A1:5662.5 UCP2:5662.5     |
| CD4-positive, Alpha-beta T Cell Activati  | -0.05475597 | 5         | 7.126e-01 | 9.886e-01 | PKG7:605 STOML2:7003.5 TNFSF4:7003.5 RSAD2:7003.5 HMR1:7003.5 NA NA                                      |
| CD40 Signaling Pathway (GO:0023035)       | 0.11701448  | 7         | 2.837e-01 | 9.886e-01 | CD86:62 CD40L:5662.5 ITGB1:5662.5 PHB2:5662.5 TNIP2:5662.5 ITGAS:5662.5 RNF31:5662.5                     |
| CENP-A Containing Chromatin Assembly (GO  | 0.04563737  | 5         | 7.236e-01 | 9.886e-01 | MIS18A:728 CENP1:5662.5 OIP5:5662.5 NASP:5662.5 HUAP:5662.5 JHURP:11122 NA NA                            |
| COP1 Coating Of Golgi Vesicle (GO:004820  | 0.05374924  | 2         | 7.923e-01 | 9.886e-01 | GBF1:5662.5 ARFGAP3:5662.5 NA NA NA NA NA                                                                |
| COP1-mediated Vesicle Budding (GO:0035964 | 0.05374924  | 2         | 7.923e-01 | 9.886e-01 | GBF1:5662.5 ARFGAP3:5662.5 NA NA NA NA NA                                                                |
| COPII Vesicle Coating (GO:0048208)        | -0.04364555 | 14        | 5.719e-01 | 9.886e-01 | KLHL12:438 TRAPPCC1:1323 TRAPPCC12:2028 TRAPPCC3:7003.5 TRAPPCC2:7003.5 TRAPPCC4:7003.5 TRAPPCC11:7003.5 |
| CRD-mediated mRNA Stabilization (GO:0070  | -0.14997459 | 5         | 2.455e-01 | 9.886e-01 | YBX1:306 IGF2BP1:841 DHX9:7003.5 PAPBC1:7003.5 HNRNP:7003.5 NA NA NA NA                                  |
| DNA Alkylation (GO:0006305)               | 0.07697699  | 15        | 3.021e-01 | 9.886e-01 | DNMT3B:57 BAZ2A:397 MTRFR:5662.5 HELLS:5662.5 ATGU:5662.5 DNMT3A:5662.5 EHMT1:5662.5                     |
| DNA Catabolic Process (GO:0006308)        | -0.04419696 | 17        | 5.282e-01 | 9.886e-01 | DICER1:391 REXO4:445 DFFR:881 DNASEL1:3900 DNASEL1:2106 ENDOG:7003.5 FBH1:7003.5                         |
| DNA Conformation Change (GO:0071103)      | -0.04860193 | 9         | 6.137e-01 | 9.886e-01 | HMG3B:599 TOP3A:1756 TOP2B:7003.5 HMGBl:7003.5 TOP2A:7003.5 TOP3B:7003.5 ERCC3:7003.5                    |

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

| Geneset                                    | stat         | num.genes | pval      | p.adj     | gene.vals                                                                                |
|--------------------------------------------|--------------|-----------|-----------|-----------|------------------------------------------------------------------------------------------|
| NIKOLSKY_BREAST_CANCER_Q0I2_Q13_AMPLICO    | -0.146877141 | 100       | 4.038e-07 | 2.615e-03 | COL20A1:29 FAM217B:76 MTG2:103 HELZ:139 CDH26:183 ZMYND8:197 SLC04A1:251                 |
| NIKOLSKY_BREAST_CANCER_IQ2I_AMPLICON       | 0.230933275  | 27        | 1.972e-05 | 6.384e-02 | ADAM15:62 PMW:204 RORC:206 MTX1:436 TDRKH:504 ZBTB7B:522 FDP5:557                        |
| KEGG_ALLOGRAFT_REJECTION                   | 0.230933275  | 21        | 2.496e-04 | 5.388e-01 | CD86:62 IFNG:93 FASLG:99 IL12B:143 HLA-DQB:232 CD80:296 CD28:361                         |
| PEREZ_TP53_TARGETS                         | -0.036632011 | 812       | 4.740e-04 | 6.139e-01 | GPR143:434 CEMIP:6 APC2:25 MDK2B:26 UNKL:55 SMTNL2:67 EME2:70                            |
| MIKKELSEN_MCIV_HCP_WITH_H3K27ME3           | -0.056870758 | 333       | 3.910e-04 | 6.139e-01 | KCNH2:1 CHRN2:13 CHRNA3:35 HTRM:27 H7 ZBTB16:69 HTR1:625 TDR6:155                        |
| WP_EFFECTS_OF_NITRIC_OXIDE                 | -0.035603951 | 6         | 7.912e-04 | 8.540e-01 | NOS3:244 MB3A:404 AOX1:1500 H2AF16:108 NOS1:2017 NOS2:2233 NA                            |
| CHARAFE_BREAST_CANCER_LUMINAL_VS_MESENCH   | -0.058303326 | 319       | 1.018e-03 | 9.419e-01 | EP58L1:52 SEMA4:104 FAM83H:117 MDPH:152 KRT19:198 CAPN8:238 GTG6:241                     |
| SHARMA_ASTROCYTOMA_WITH_NF1_SYNDROM        | -0.086348307 | 3         | 6.045e-01 | 9.986e-01 | CLEC3B:1708 ADGRV1:7003.5 SLC1A3:7003.5 NA NA NA NA NA                                   |
| SOGA_COLORECTAL_CANCER_MYC_CN              | 0.038979579  | 56        | 3.134e-01 | 9.986e-01 | ACO2:74 SULT1A2:323 SLC44A1:614 SULT1A4:684 PLPPI:865 B3GNT5:994 SL4C4A:5662.5           |
| SOGA_COLORECTAL_CANCER_MYC_CN              | 0.028444909  | 63        | 4.353e-01 | 9.986e-01 | GPI:398 UMP5:507 SLC7A6:554 POLR3K:734 DTYMK:985 PAICS:5662.5 CAD:5662.5                 |
| CORONEL_RFX7_DIRECT_TARGETS_UP             | 0.014331788  | 23        | 8.120e-01 | 9.986e-01 | XRCC1:905 TSPYL2:1000 ABAT:5662.5 KIF9:5662.5 DDT3:5662.5 CREBZF:5662.5 NA NA            |
| FOROUTAN_TGFB_EMT_UP                       | -0.008841825 | 138       | 7.205e-01 | 9.986e-01 | SACS:64 MMP9:328 TUF1:407 EML14:644 ARFGAP1:582 PLEK2:594 LARP6:634                      |
| FOROUTAN_TGFB_EMT_DOWN                     | -0.060006315 | 81        | 6.224e-02 | 9.986e-01 | CAVIN2:69 ERMP1:186 KRT19:198 SLC04A1:251 LAMA5:295 JAG2:319 KRT15:330                   |
| FOROUTAN_PRODRAK_TGFB_EMT_UP               | -0.017890028 | 132       | 4.788e-01 | 9.986e-01 | SACS:64 MMP9:328 TUF1:407 EML14:644 ARFGAP1:582 PLEK2:594 LARP6:634                      |
| FOROUTAN_PRODRAK_TGFB_EMT_DOWN             | -0.052924699 | 64        | 1.435e-01 | 9.986e-01 | CAVIN2:69 ERMP1:186 KRT19:198 SLC04A1:251 LAMA5:295 JAG2:319 SYBU:742                    |
| FOROUTAN_INTEGRATED_TGFB_EMT_UP            | 0.008761891  | 82        | 7.841e-01 | 9.986e-01 | COL1A1:148 VGLL3:427 BMPR2:512 COL6A3:646 GALNT10:676 SLC26A2:5662.5 JAG1:5662.5         |
| FOROUTAN_INTEGRATED_TGFB_EMT_DOWN          | -0.048390018 | 57        | 2.067e-01 | 9.986e-01 | CAVIN2:69 ERMP1:186 KRT19:198 KRT15:330 ATP8B1:805 BIRC3:885 ELF3:896                    |
| GLASS_IGF2BP1_CLIP_TARGETS_KNOCKDOWN_DN    | 0.001233343  | 85        | 9.688e-01 | 9.986e-01 | NCEH1:720 CD2AP:570 SRPK1:882 N4BP2:928 SHMT2:5662.5 HSD1B6:5662.5 HNRNP:5662.5          |
| HOUNKPE_HOUSEKEEPING_GENES                 | 0.001427951  | 694       | 9.939e-01 | 9.986e-01 | RNF168:118 MBD1:58 NARSL:194 CCAR2:126 IP08:174 PCDOT:178 MRPS30:265                     |
| IBRAHIM_NRF1_UP                            | 0.009226787  | 275       | 6.002e-01 | 9.986e-01 | DNMT3B:57 MBD1:58 NCKAP1:92 NARSL:94 SCPEP1:149 FUBP1:152 SUPV3L1:224                    |
| IBRAHIM_NRF2_UP                            | 0.019770601  | 358       | 2.016e-01 | 9.986e-01 | NCKAP1:92 NARSL:94 TYW3:131 SUPV3L1:224 ZNF189:253 MRPS30:265 CLU:279                    |
| IBRAHIM_NRF3_UP                            | 0.053253480  | 2         | 7.942e-01 | 9.986e-01 | HMOX1:5662.5 RPN2:5662.5 NA NA NA NA NA                                                  |
| IBRAHIM_NRF1_DOWN                          | -0.031722755 | 94        | 2.885e-01 | 9.986e-01 | TMEM25:72 KCNH3:95 NPHP4:116 LIG1:243 PID1:275 PBXIP1:569 PCLO:628                       |
| IBRAHIM_NRF2_DOWN                          | -0.052893290 | 109       | 5.686e-02 | 9.986e-01 | IQGAP3:34 CDK10:102 ENGR1:226 LIG1:243 PPP1R3A:266 PID1:275 RHPN1:492                    |
| IBRAHIM_NRF3_DOWN                          | 0.0056126754 | 11        | 5.560e-01 | 9.986e-01 | ARHGAP23:700 SMAP1:5662.5 NGS:5662.5 PLAT:5662.5 SEMA6:5662.5 HECTD3:5662.5 IER2:5662.5  |
| SEAVEY_EPITHELIOID_HEMANGIOENDOTHELIOMA    | 0.040657707  | 67        | 2.503e-01 | 9.986e-01 | C17orf58:205 TNFSF15:272 CDC42CP5:564 NCEH1:720 MYL2:758 PLCD3:5662.5 SERPINEL1:5662.5   |
| SENATZ_GBM_PROTEASOME_INHIBITION_RESPON    | 0.002669999  | 31        | 6.934e-01 | 9.986e-01 | ELP6:978 ASNS:5662.5 MLF2:5662.5 GPATCH12:5662.5 PNRC1:5662.5 SOX4:5662.5 SESN2:5662.5   |
| BLANCO_MELO_SARS_COV_1_INFECTION_MCR5_CE   | 0.053275038  | 7         | 2.255e-01 | 9.986e-01 | KIF14:5662.5 CENPE:5662.5 ASPM:5662.5 ESM1:5662.5 KIF20B:5662.5 BRCA2:5662.5 SMC4:5662.5 |
| BLANCO_MELO_SARS_COV_1_INFECTION_MCR5_CELL | 0.173739275  | 4         | 2.288e-01 | 9.986e-01 | HSD17B2:52 CYTL1:446 FGF4:5662.5 RASL12:10386 NA NA NA                                   |
| BLANCO_MELO_MERS_COV_INFECTION_MCR5_CELL   | -0.022826641 | 187       | 2.832e-01 | 9.986e-01 | SUSD4:68 KCNH3:95 RBM15:184 CDKN2AP:144 ATP2A1:595 FHDC1:650 MED26:743                   |
| BLANCO_MELO_MERS_COV_INFECTION_MCR5_CELL   | 0.103302480  | 20        | 3.488e-02 | 9.986e-01 | HSD17B2:52 REK1B:89 HSD11B1:428 CYP11A4:629 SLC25A1:581 PNMA2:681 AIBG:5662.5            |
| BLANCO_MELO_INFLUENZA_A_INFECTION_A594_C   | 0.056309102  | 26        | 3.205e-01 | 9.986e-01 | ACCL2:69 PLAUR:687 ERERG:736 MRPL2:419 CNA2:5662.5 FGFBP1:5662.5 ANKRD1:5662.5           |
| BLANCO_MELO_INFLUENZA_A_INFECTION_A594_C   | -0.043790195 | 78        | 1.817e-01 | 9.986e-01 | PXCR:25 OPLAH:364 OVC2A:509 TONSLB:99A SPAT2:5662.5 T7TBC1D1:676 CHPF2:848               |
| BLANCO_MELO_HUMAN_PARAINFLUENZA_VIRUS_3    | 0.003666834  | 132       | 8.010e-01 | 9.986e-01 | TNFSF10:36 MYPN:41 STAT2:129 OAS2:281 PLEKHA4:290 HCAAR2:299 CXCL11:372                  |
| BLANCO_MELO_HUMAN_PARAINFLUENZA_VIRUS_3    | 0.039064052  | 8         | 9.020e-01 | 9.986e-01 | PROD2:40 C5C5:79 SERPINA1:639 ECEB1:225 CAMSAP3:43 CA9:467 TUBBA4A:619                   |
| BLANCO_MELO_RESPIRATORY_SYNCYTIAL_VIRUS_   | 0.001053896  | 194       | 5.958e-01 | 9.986e-01 | TNFSF10:36 STAT2:129 PLEKHA4:290 HCAAR2:299 TNFRSF9:312 TNF:363 CXCL11:372               |
| BLANCO_MELO_RESPIRATORY_SYNCYTIAL_VIRUS_   | -0.044303487 | 83        | 1.635e-01 | 9.986e-01 | MCFZL:317 PAD2:1348 MB:354 DACT2:430 HLA-DMB:453 SYNGR4:534 HGBX13:640                   |
| BLANCO_MELO_COVID19_SARS_COV_2_INFECTION   | -0.038275814 | 59        | 3.096e-01 | 9.986e-01 | N4BP3:176 DNAT1:7360 RASGRP2:389 SDRP16C5:395 INHBE:622 BCL2A1:759 PCDH1:854             |
| BLANCO_MELO_COVID19_SARS_COV_2_INFECTION   | -0.086149760 | 51        | 3.344e-02 | 9.986e-01 | KLHDCA7:195 VLI1:326 SLC25A10:324 DACT2:430 HLA-DMB:453 SYNGR4:534 RADIL:664             |
| BLANCO_MELO_COVID19_SARS_COV_2_INFECTION   | 0.007984507  | 240       | 6.714e-01 | 9.986e-01 | TNFSF10:36 HDAC9:43 STAT2:129 OAS2:281 NEDD4:300 TNFRSF9:312 TNF:363                     |

DisGeNET Top pathways by non-permutation

|                                          |             |     |           |           |                                                                                      |
|------------------------------------------|-------------|-----|-----------|-----------|--------------------------------------------------------------------------------------|
| Albinism, Ocular                         | -0.08184242 | 22  | 1.841e-01 | 9.739e-01 | GPR143:4 DTNBP1:60 HPS4:469 LYST:730 AP3D1:1514 SLCA5A:1726 TYRP1:7003.5             |
| Albinism, Oculocutaneous                 | 0.03076540  | 27  | 5.803e-01 | 9.739e-01 | PMEL:346 CTLA4:566 CSF3:5662.5 NRAS:5662.5 KIT:5662.5 PIK3CA:5662.5 CDKN2A:5662.5    |
| Alkalosis, Respiratory                   | 0.14732627  | 4   | 3.075e-01 | 9.739e-01 | ASL904 CA5A:5662.5 CPS1:5662.5 OTC:5662.5 NA NA NA                                   |
| Alopecia, Male Pattern                   | 0.06720986  | 26  | 2.358e-01 | 9.739e-01 | HDAC9:43 SRD5A2:201 SUPV3L1:224 WNT4:516 NFE2L2:578 MTHFR:5662.5 IGF1:5662.5         |
| alpha-1-Antitrypsin Deficiency, Autosoma | 0.08438347  | 18  | 2.153e-01 | 9.739e-01 | ACE:302 TNF:363 NFE2L2:578 ELANE:625 A2M:920 CAT:5662.5 SFKN1:5662.5                 |
| ALPHA-THALASSEMIA/MENTAL RETARDATION SYN | 0.04468955  | 14  | 5.627e-01 | 9.739e-01 | DAXX:594 PML:5662.5 TP53:5662.5 ALK:5662.5 SMARCA1:5662.5 ATRX:5662.5 SMARCA2:5662.5 |
| Alzheimer's Disease, Global Onset        | 0.03006435  | 58  | 4.290e-01 | 9.739e-01 | CLU:279 ACE:302 TNF:363 MPO:391 TFAM:413 CHRNA7:415 CD2AP:750                        |
| Alzheimer Disease, Early Onset           | 0.04317460  | 80  | 1.827e-01 | 9.739e-01 | CLU:279 ACE:302 MEVY:305 TNF:363 MPO:391 ULNQLN1:395 TFAM:413                        |
| Alzheimer Disease, Late Onset            | 0.02950459  | 185 | 1.684e-01 | 9.739e-01 | TET1:24 DNMT3B:57 LRPI:175 DNMT1:193 CLU:279 ACE:302 HPDGS:330                       |
| Amaurosis congenita of Leber, type 1     | 0.02538800  | 41  | 5.741e-01 | 9.739e-01 | NFE2L2:578 CXCL6:28 HD12:725 MEFC2:761 PTPRR:780 CEP290:952 ERBB2:5662.5             |
| Ambiguous genitalia, male                | 0.05775740  | 6   | 6.242e-01 | 9.739e-01 | SRD5A2:201 WT1:5662.5 CYP11A1:5662.5 POR:5662.5 HD53B2:5662.5 MKS1:10758 NA          |

KeggReactome Top pathways by non-permutation

| Geneset                                  | stat        | num.genes | pval      | p.adj     | gene.vals                                                                          |
|------------------------------------------|-------------|-----------|-----------|-----------|------------------------------------------------------------------------------------|
| REACTOME_THE_CITRIC_ACID_TCA_CYCLE_AND_R | 0.14843331  | 146       | 7.201e-10 | 1.298e-06 | NDUFS5:22 NDUFS3:22 ATP5PD:29 NDUFB5:50 NDUFA12:63 NDUFA11:69 NDUFB3:71            |
| REACTOME_SENSORY_PERCEPTION              | -0.09711505 | 291       | 1.659e-08 | 1.496e-05 | PCDH15:2 RIPOR2:80 CTBP2:96 CALHM1:108 OR10S1:109 TRPM5:140 OR6Y1:177              |
| REACTOME_RESPIRATORY_ELECTRON_TRANSPORT_ | 0.16102289  | 99        | 3.412e-08 | 1.538e-05 | NDUFS5:22 NDUFS3:22 ATP5PD:29 NDUFB5:50 NDUFA12:63 NDUFA11:69 NDUFB3:71            |
| REACTOME_OLFACTORY_SIGNALING_PATHWAY     | -0.15529358 | 105       | 4.264e-08 | 1.538e-05 | OR10S1:109 OR6Y1:177 OR13D1:285 OR52E8:323 OR6B1:35 OR10J5:357 OR5111:398          |
| KEGG_OXIDATIVE_PHOSPHORYLATION           | 0.16588700  | 92        | 4.180e-08 | 1.538e-05 | NDUFS5:22 NDUFS3:22 ATP5PD:29 NDUFB5:50 NDUFA11:69 NDUFB3:71 NDUFB8:117            |
| REACTOME_COMPLEX_I_BIOGENESIS            | 0.21983167  | 48        | 1.429e-07 | 4.295e-05 | NDUFS5:22 NDUFS3:22 NDUFB5:50 NDUFA12:63 NDUFA11:69 NDUFB3:71 NDUFB8:117           |
| REACTOME_RESPIRATORY_ELECTRON_TRANSPORT  | 0.16255479  | 79        | 6.276e-07 | 1.617e-04 | NDUFS5:22 NDUFS3:22 NDUFB5:50 NDUFA12:63 NDUFA11:69 NDUFB3:71 LRPPRC:113           |
| KEGG_HUNTINGTONS_DISEASE                 | 0.12474588  | 128       | 1.220e-06 | 2.749e-04 | NDUFS5:22 NDUFS3:22 ATP5PD:29 NDUFB5:50 NDUFB3:71 TFAM:103 NDUFB8:117              |
| KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACT | 0.09657501  | 205       | 2.103e-06 | 4.212e-04 | ILTA:47 GH2:62 IFNG:73 TNFSF8:87 CXCR2:168 L16ST:183 TNFSF10:201                   |
| KEGG_OLFACTORY_TRANSDUCTION              | -0.12153772 | 119       | 5.072e-06 | 9.144e-04 | OR10S1:109 OR6Y1:177 OR13D1:285 OR52E8:323 OR6B1:35 OR10J5:357 CAMK2B:373          |
| REACTOME_CLASS_A_1_RHOODPSIN_LIKE_RECEPT | -0.07735958 | 294       | 6.232e-06 | 1.021e-03 | GPR143:33 HTR1D:15 SSTR4:53 CATR2:170 HTR6:175 PEUF:255 CCR9:259                   |
| KEGG_PARKINSONS_DISEASE                  | 0.13810130  | 89        | 7.110e-06 | 1.068e-03 | NDUFS5:22 NDUFS3:22 ATP5PD:29 NDUFB5:50 NDUFB3:71 NDUFB8:117 NDUFA8:314            |
| REACTOME_SPERM_MOTILITY_AND_TAXES        | 0.06551278  | 8         | 6.863e-05 | 9.518e-03 | CATSPERB:4 CATSPER3:60 CATSPER4:186 CATSPERG:20 CATSPER2:37 CATSPERD:55 KCNU1:1291 |
| REACTOME_METABOLISM_OF_STEROID_HORMONES  | 0.20501480  | 30        | 1.031e-04 | 1.325e-02 | CGA:99 HSD11B1:198 HSD17B2:460 CYP17A1:479 SRD5A3:626 HSD3B2:649 AKR1B1:731        |
| REACTOME_AMINE_LIGAND_BINDING_RECEPTORS  | -0.17474048 | 41        | 1.103e-04 | 1.325e-02 | GPR143:33 HTR1D:15 HTR6:175 ADRA2C:387 TAAR6:425 ADRI1:874 ADRA2A:908              |
| REACTOME_GLUCCORTICOID_BIOSYNTHESIS      | 0.43134178  | 6         | 2.536e-04 | 2.689e-02 | HSD11B1:198 CYP17A1:479 HSD3B2:649 HSD11B2:337 CYP21A2:1918 SERPINA6:1940 NA       |
| REACTOME_FORMATION_OF_THE_CORNIFIED_ENVE | -0.1278077  | 76        | 2.504e-04 | 2.689e-02 | EVPL:97 KRT78:157 PPL:186 KRT80:296 DSG2:300 PKP3:442 KRT85:514                    |
| REACTOME_ADAPTIVE_IMMUNE_SYSTEM          | 0.04468939  | 580       | 3.117e-04 | 2.810e-02 | FBXW12:8 UBR2:17 COL2A1:28 HSPA5:64 KLHL5:104 S100A8:116 CD90:118                  |
| REACTOME_SIGNALING_BY_GPCR               | -0.0444705  | 595       | 2.833e-04 | 2.810e-02 | GPR143:33 HTR1D:15 SSTR4:53 PKR3:78 ARHGFE11:87 HSD3B2:649 HTR6:175                |
| KEGG_STEROID_HORMONE_BIOSYNTHESIS        | -0.19733558 | 28        | 3.046e-04 | 2.810e-02 | HSD11B1:198 AKR1D1:222 HSD17B2:460 CYP17A1:479 SRD5A3:626 HSD3B2:649 HSD17B6:830   |
| KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERAC | -0.06502510 | 260       | 3.454e-04 | 2.966e-02 | GRIN3B:12 HSD17D:15 SSTR4:53 TACR2:170 SEHL:195 P2RX2:205 CHRN2:252                |
| REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY | 0.04295700  | 555       | 6.869e-04 | 5.170e-02 | ILTA:47 GH2:62 IFNG:73 TNFSF8:87 TNF:363 CD80:118 CD86:119                         |
| REACTOME_INTERCONVERSION_OF_NUCLEOTIDE_D | 0.18586595  | 28        | 6.703e-04 | 5.170e-02 | AK9:46 AK6:298 CTPS2:435 NUDT13:473 DTYMK:654 NME4:1275 TXN:1435                   |
| REACTOME_GPCR_LIGAND_BINDING             | -0.05098919 | 386       | 6.882e-04 | 5.170e-02 | GPR143:33 HTR1D:15 SSTR4:53 TACR2:170 HTR6:175 TSAR2:131 TPN:225                   |
| REACTOME_ANDROGEN_BIOSYNTHESIS           | 0.32184595  | 9         | 8.292e-04 | 5.816e-02 | CGA:99 CYP17A1:479 SRD5A3:626 HSD3B2:649 HSD17B3:941 LH:1690 SRD5A2:2441           |
| REACTOME_SIGNALING_BY_INTERLEUKINS       | 0.05048379  | 381       | 8.386e-04 | 5.816e-02 | IFNG:73 CD80:118 CD86:119 PIK3R3:164 L16ST:183 ATEL1:226 CAT:528                   |
| KEGG_ALZHEIMERS_DISEASE                  | 0.08658605  | 125       | 8.715e-04 | 5.820e-02 | NDUFS5:22 NDUFS3:22 ATP5PD:29 NDUFB5:50 PPP3C:67 NDUFB3:71 NDUFB8:117              |
| REACTOME_EPIGENETIC_REGULATION_OF_GENE_E | 0.09727688  | 96        | 1.025e-03 | 6.440e-02 | DNMT3B:31 MYOIC5:51 DEK:83 TET1:102 TET2:118 H3-3A:233 DNMT1:235                   |
| REACTOME_REACTIONS_SPECIFIC_TO_THE_COMPL | 0.33499411  | 8         | 1.036e-03 | 6.440e-02 | CGA:99 MGAT:576 MAN2A1:753 LHB:1690 CHST10:1871 MAN2A2:2996 FUT8:3093              |
| REACTOME_G_ALPHA_12_13_SIGNALLING_EVENTS | -0.11888361 | 62        | 1.232e-03 | 6.912e-02 | ARHGFE11:87 ARHGEF15:257 OBCSN:309 TRO:402 TAMI:621 ARHGEF10L:622 FGD1:820         |
| REACTOME_INTERLEUKIN_10_SIGNALING        | 0.15818553  | 35        | 1.215e-03 | 6.912e-02 | CD80:118 CD86:119 CXCL2:328 CSF3:447 CXCL10:522 IL5:47 IL28:58                     |
| REACTOME_KERATINIZATION                  | -0.10263306 | 83        | 1.265e-03 | 6.912e-02 | EVPL:97 KRT78:157 PPL:186 KRT80:296 DSG2:300 PKP3:442 KRT85:514                    |
| REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION | 0.03346534  | 867       | 1.182e-03 | 6.912e-02 | IFNG:73 DEK:83 PRDX1:89 CGA:99 PRMT5:111 LRPPRC:113 TPX2:115                       |
| REACTOME_PIWI_INTERACTING_RNA_PIWI_BIOG  | -0.22454617 | 17        | 1.356e-03 | 7.189e-02 | MOV10L1:56 TDG2:111 TDRD6:328 TDRK4:417 PIV1:1841 POLR2A:935 DDX4:970              |
| REACTOME_TNFR2_NON_CANONICAL_NF_KB_PATHW | 0.09505404  | 78        | 2.439e-03 | 1.257e-01 | ATL:47 TNFSF8:87 KCTD:228 PSMB2:270 NKFB2:618 TNF:661 FASLG:949                    |
| REACTOME_NGF_STIMULATED_TRANSCRIPTION    | 0.15617023  | 31        | 2.641e-03 | 1.323e-01 | TF1:226 PTHL7:638 S3GL3:843 JUNB:954 JUND:1065 TRAF1:1261 NAB1:1373                |
| REACTOME_METABOLIC_DISORDERS_OF_BIOLOGIC | 0.17485677  | 30        | 2.855e-03 | 1.391e-01 | CYP17A1:479 CYP26C1:17 LSCD3:651 CYP19A1:1423 FMOS:156 CYP27A1:1605 CYP27B1:1652   |
| REACTOME_MINERALOCORTICOID_BIOSYNTHESIS  | 0.42575367  | 4         | 3.189e-03 | 1.513e-01 | CGA:99 HSD3B2:649 LH:1690 CYP21A2:1918 NA NA                                       |
| REACTOME_CRISTAE_FORMATION               | 0.16914177  | 25        | 3.441e-03 | 1.591e-01 | ATP5PD:29 ATP5MG:770 DMACL2:1097 ATP5MF:1519 ATP5PB:1675 DNAJC11:2191 ATP5F1A:2403 |
| REACTOME_HDR_THROUGH_HOMOLOGOUS_RECOMBIN | -0.10658079 | 62        | 3.769e-03 | 1.699e-01 | POLD2:41 EMEZ:225 POLY:413 HUS1:428 RTEL:147 MRE11:660 EME1:892                    |

MGI\_Mammalian\_Phenotype\_Level\_4 Top pathways by non-permutation